

Card: expression-distribution

card_id: expression-distribution · version: 1.0.0 · generation: modern (rules externalized) · schema_version: 1

Across ~1500 DepMap cell lines, what is the expression distribution of the target, and is it lineage-stratified?

At a glance

```
depmap-consortium-26q1 (DepMap 26Q1 cell-line RNA, release-pinned)
|   method: depmap-expression-distribution
v
summary_fields (percentiles, fractions, per-lineage stats)
|   classify
v
expression_class = one of { broadly_high | broadly_moderate | lineage_restricted | broadly_low | data_unavailable }
|   5 rules (intracellular-intrinsic axis)
v
per-modality signal -> small_molecule / degrader
```

Data source

product_id	release_pin	What it is
depmap-consortium-26q1	{release_pin}	DepMap 26Q1 OmicsExpressionTPMLogp1HumanProteinCodingGenes.csv — p

Read is manifest-mediated (no bare S3 paths) — provenance + release-pin are automatic.

Method

methods.call	args	Computes
depmap-expression-distribution	target, release_pin, expressed_threshold	Panel expression distribution: perce

Emitted evidence (summary_fields)

Field	Notes
n_cell_lines_evaluated	panel size (drives the low-coverage wa
median_log2tpm_panel, p25/p75/p5/p95_log2tpm_panel, log2tpm_iqr	
fraction_expressed	log2(TPM+1) ≥ 1.0
fraction_highly_expressed	log2(TPM+1) ≥ 5.0
fraction_not_expressed	log2(TPM+1) < 1.0
expression_class	DEPMAP-convention categorical — driv
n_lineages_evaluated, per_lineage_stats, n_lineage_restricted_lineages	lineage stratification

Categorical vocabulary — expression_class (the value set rules key off): broadly_high (>70% highly expressed) · broadly_moderate (>70% expressed, <30% highly) · lineage_restricted (10-70% expressed, stratified) · broadly_low (<30% expressed) · data_unavailable

Plots

id	type	preferred surfaces
density_expression	density histogram + KDE (threshold refs at 1.0 / 5.0)	markdown, ppt, decision_men
lineage_strip_expression	per-lineage strip plot (n≥5, ordered by median)	markdown, ppt
waterfall_expression	ranked per-cell-line waterfall	markdown, ppt

plot_data: per_cell_line_expression_with_lineage_tags

Interpretation rules

Interpretation lives in interpretation-rules/intracellular-intrinsic.rules.yaml (not the card). Each rule fires on an expression_class value → per-modality signal.

rule_id	when expression_class =	small_molecule	degrader	dominant
expression-broadly-high-supportive	broadly_high	supportive	supportive	
expression-lineage-restricted-supportive	lineage_restricted	supportive	supportive	
expression-broadly-moderate-neutral	broadly_moderate	neutral	neutral	
expression-broadly-low-degrader-killer	broadly_low	—	killer	
expression-data-unavailable-insufficient	data_unavailable	insufficient	insufficient	

Signal vocabulary: killer / supportive / neutral / insufficient, per modality.

Gating, thresholds & warnings

- applies_when: context.data_sources.depmap_expression_available == true
- thresholds: expressed_threshold_log2tpm=1.0 · highly_expressed_threshold_log2tpm=5.0 · broadly_expressed_f · broadly_high_fraction_threshold=0.30 · lineage_restricted_min/max_fraction=0.10/0.70 · min_lineage_size=5
- warning: low_panel_expression_coverage — if n_cell_lines_evaluated < 500 → "panel coverage unusually low; matrix subset may be biased" (→ passed_with_warnings)

Caveats (verbatim)

- Cell-line expression does not match tumor expression directly; DepMap lines are immortalized cultures biased toward proliferation programs.
- Expression class is descriptive — therapeutic interpretation lives in interpretation-rules/.
- Lineage-restricted expression in DepMap is informative, but the gold-standard normal-tissue safety signal comes from GTEx via the lineage-restriction-evidence card.

Used by

tumor-presence (leaf) · target-profile (composer, via tumor-presence)